

Data for Policy Analysis and Management

CLASS 6: LINEAR REGRESSION
CONTINUATION

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Agenda

1. Linear Regression Extensions
2. Multiple Linear Regression
3. Spurious Relationships
4. Categorical Variables
5. Logistic Regression
6. Interaction Effects
7. Small Group Activity
8. Lab: Multiple Linear Regression

Why Extend Linear Regression?

- Simple linear regression models the relationship between one predictor and one outcome.
- But most policy questions involve multiple factors acting simultaneously, requiring a **more powerful approach**.

Benefits of Multiple Regression

- Model several predictors simultaneously rather than one at a time.
- Separate each variable's independent contribution to the outcome.
- Control for confounding variables to reduce bias.
- Estimate adjusted relationships holding other factors constant.
- Improve prediction accuracy and model realism.

Does grant receipt raise wages after controlling for education, experience, and provider size?

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Review: Multiple Linear Regression

Model Formula

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon$$

Each coefficient estimates the relationship with Y holding all other variables constant.

Y

Dependent variable: the outcome being explained or predicted.

$X_1 \dots X_k$

Explanatory variables: the predictors included in the model (X_1 through X_k).

$\beta_0, \beta_1 \dots \beta_k$

β_0 = intercept; $\beta_1 \dots \beta_k$ = regression coefficients; ε = random error term.

Statistical Control

Statistical control means comparing units that are similar on key characteristics, so we can isolate the association of interest.

Variables commonly controlled for:

- Education level
- Years of experience
- Provider or organization size
- Neighborhood context

Purpose: By holding these factors constant, we remove their influence and observe the relationship between our predictor and outcome more clearly.

- Statistical control does not eliminate all bias, but it substantially clarifies the associations we estimate.

Example: To assess the impact of a public program on wages, we compare staff who share similar education and experience levels. This ensures any observed wage difference is more likely attributable to grant participation, not to background characteristics.

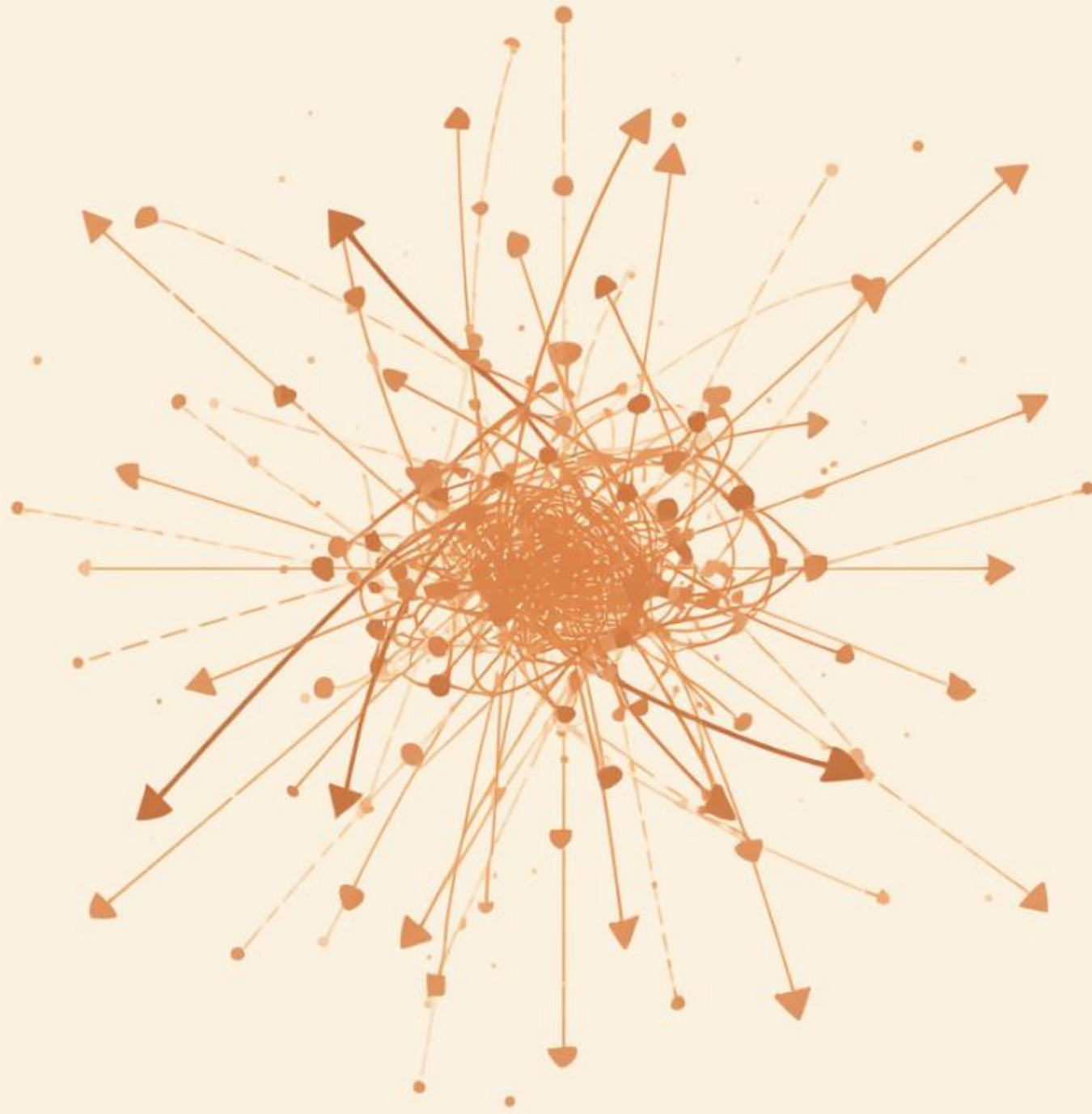
Why Statistical Control Matters

The Risk of Ignoring Confounders

- Many variables influence policy outcomes at the same time.
- Failing to account for them can distort our understanding of what is really driving the results we observe.

Key Points

- Multiple variables simultaneously influence outcomes.
- Ignoring relevant variables introduces bias and misleading conclusions.
- Observed differences may reflect other variables, not the factor of interest.
- Misattributing effects can lead to poor policy decisions!
- Regression estimates adjusted associations, not just raw comparisons.
- Control helps isolate the true relationship between predictor and outcome.



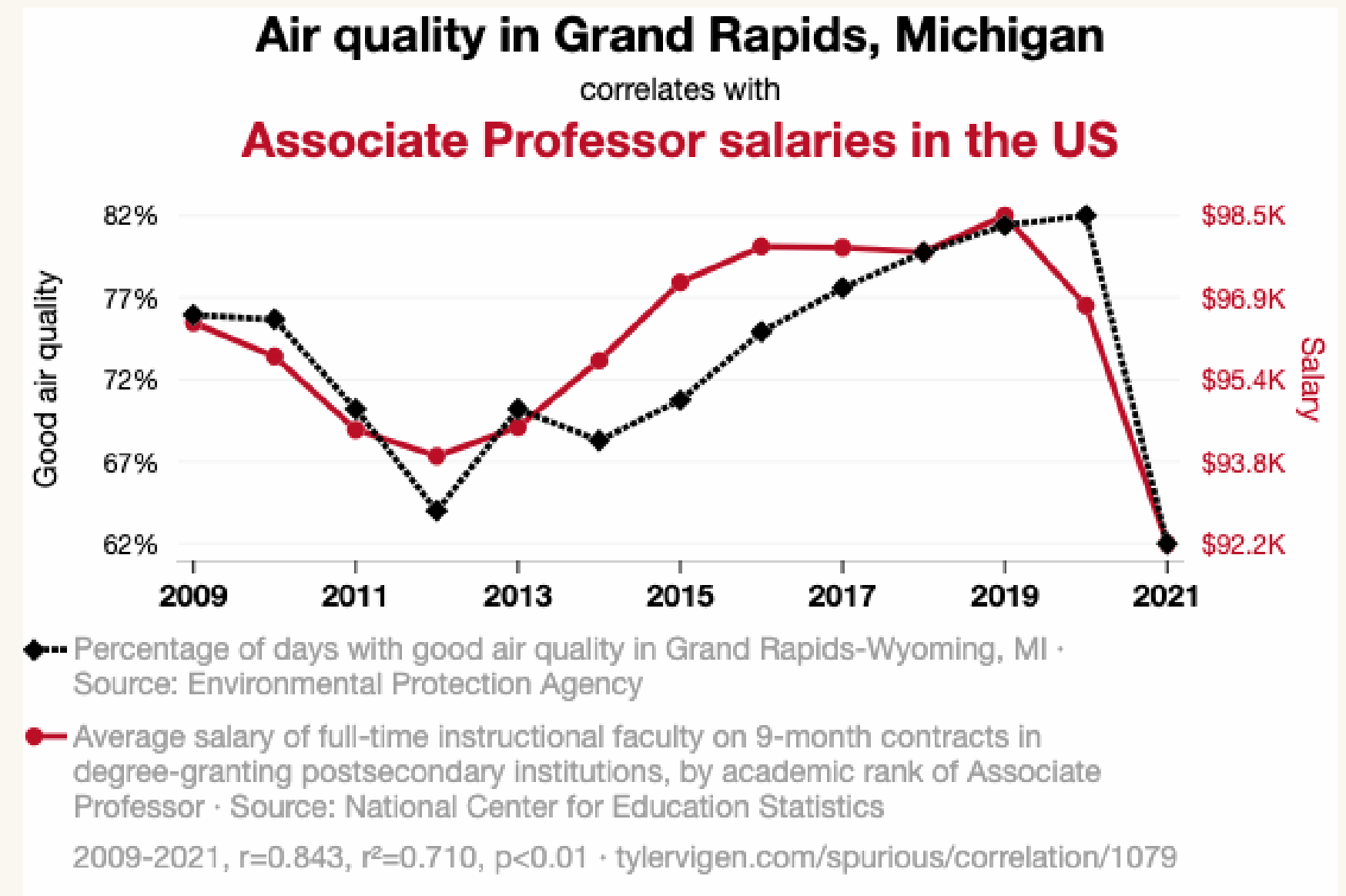
Spurious Relationships

Definition

- An apparent association between two variables caused by a third variable. This is **not a true** direct relationship.
- Controlling for the third variable may reduce or eliminate the observed association.

Key Example

- Quality air vs. Associate professor salaries in the U.S.
- Association $\neq$ Causation
- Regression helps reveal whether a relationship holds after accounting for confounding variables.



“I have 25,153 variables in my database. I compare all these variables against each other to find ones that randomly match up. That's 632,673,409 correlation calculations! This is called “data dredging.” Instead of starting with a hypothesis and testing it, I instead abused the data to see what correlations shake out. It's a dangerous way to go about analysis, because any sufficiently large dataset will yield strong correlations completely at random.”

Categorical Variables

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Key Concept

- Many policy-relevant predictors are categorical rather than continuous.
- These variables are incorporated into regression using dummy (indicator) variables: coded as 1 if a unit belongs to a category, and 0 if not.
- The coefficient on a dummy variable represents **the average difference in the outcome between that group and the reference group**, holding other variables constant.
- Common categorical predictors: gender, provider type, grant participation, rural/urban



Variables with Multiple Groups

- When a categorical variable has more than two categories, we create multiple dummy variables to capture all group differences.
- Examples of multi-category variables:
 - Education level (high school, associate, bachelor's, graduate)
 - Race/ethnicity
 - Geographic region
 - Program quality rating
- Creating Dummy Variables:
 - For a variable with k categories, create $k-1$ dummy variables. For example, education with 4 levels requires 3 dummy variables.
- Reference Group:
 - One category is omitted and serves as the reference group. All coefficients are interpreted as differences relative to this baseline category.



Interpreting Multiple Categories

Regression Table

Education Coefficients

Education Level	Coefficient
High School (ref.)	—
Associate Degree	+\$1.25/hr
Bachelor's Degree	+\$3.10/hr
Graduate Degree	+\$5.40/hr

How to Read the Results

Interpretation

Compared to high school diploma holders:

- Associate degree holders earn \$1.25/hr more on average
- Bachelor's degree holders earn \$3.10/hr more on average
- Graduate degree holders earn \$5.40/hr more on average

The intercept represents the expected hourly wage for the reference category (i.e., workers with a high school diploma) holding all other variables constant.



Choosing the Reference Group

- The reference group is the baseline category against which all other groups are compared.
- Choosing it thoughtfully improves clarity and interpretability of results.

Key Guidance

- Good reference groups: largest category, untreated group, policy baseline, or lowest education level.
- Changing the reference group changes coefficient estimates but not predicted values or overall model fit.
- All groups are still represented, only the comparison point shifts.
- Always clearly identify the reference group when reporting regression results.

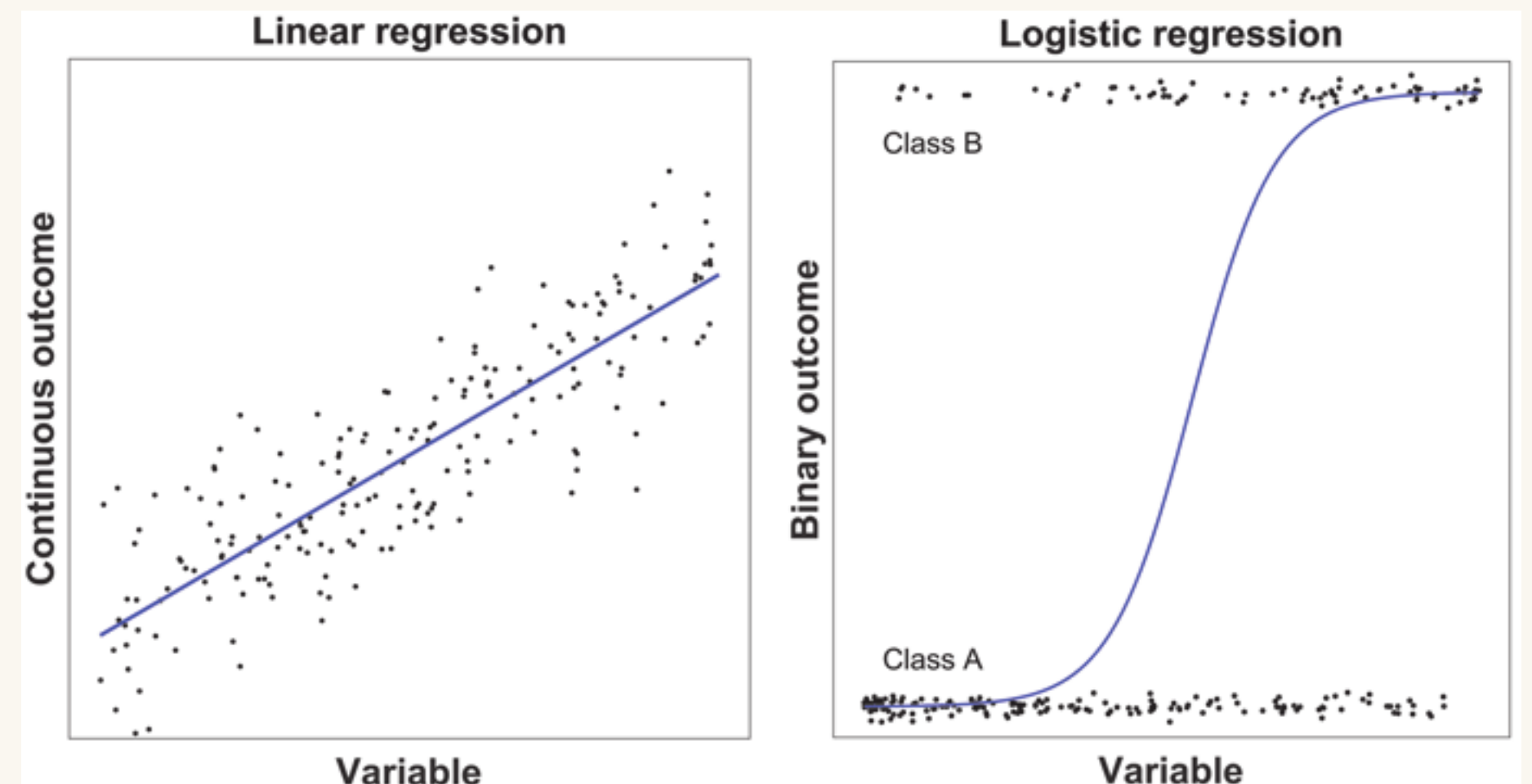
When OLS Is Not Enough

The Problem

- OLS is well-suited for continuous dependent variables.
- However, many policy outcomes are binary, they take only two values: Yes or No.
- Examples include grant receipt, provider closure, full-time employment, preschool enrollment, among others.

Problems with OLS on Binary Outcomes

- Predictions can fall outside the 0–1 range, producing impossible probability values.
- Relationship between predictors and probabilities is non-linear, violating OLS assumptions.
- Heteroskedasticity: variance is non-constant with binary outcomes.
- Solution: **Logistic regression**, designed specifically for binary dependent variables



Logistic Regression

1

Models the probability that an event occurs ($Y = 1$), such as grant receipt or provider closure.

2

Predicts log-odds rather than a direct outcome value.
Example predictors: program quality, capacity, provider type, neighborhood poverty, years operating.

3

Probabilities are constrained between 0 and 1, avoiding the out-of-range predictions that occur with OLS on binary outcomes.

Use logistic regression whenever your dependent variable is binary, it models probability, not magnitude.

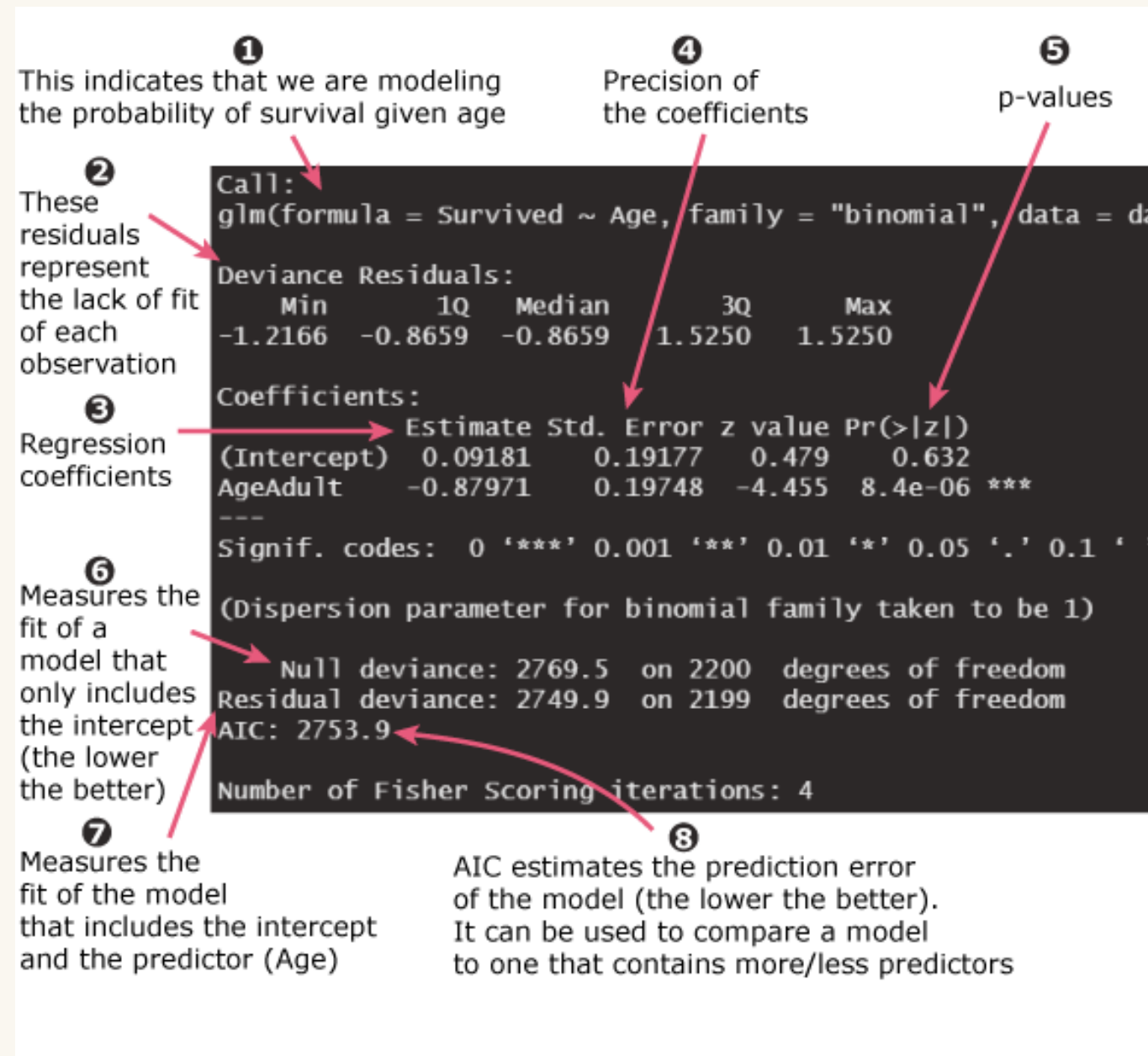
Interpreting Logistic Regression

Key Outputs & Interpretation

- Odds ratio: Change in the odds of the outcome associated with a one-unit change in a predictor.
- Predicted probability: Estimated probability that the outcome occurs for a given set of predictors.
- Marginal effect: Change in the predicted probability associated with a one-unit change in a predictor.

• Example:

- Odds ratio = 1.40 means 40% higher odds of the outcome, holding all other variables constant.
- Holding all other variables constant, providers receiving technical assistance have 40% higher odds of receiving a Smart Start Workforce Grant than providers that did not receive technical assistance.
- Odds $\neq$ probabilities, they measure the same thing differently
- Odds ratios > 1 indicate higher odds; < 1 indicate lower odds



Understanding the Coefficients

- Logistic regression coefficients are expressed in **log-odds**, which are difficult to interpret directly.
- Results are typically converted into more intuitive outputs for policy analysis (e.g., odds ratios, predicted probability, marginal effects).

Linear Regression

- Outcome type: Continuous variable (e.g., hourly wage, annual income).
- Predicts: Fitted values along a line
- Relationship shape: Linear
- Coefficients: Unit changes in Y per unit change in X.
- Estimation method: Ordinary Least Squares (OLS).
- Use when: Dependent variable is continuous.

Logistic Regression

- Outcome type: Binary variable (e.g., grant receipt, provider closure)
- Predicts: Probabilities constrained between 0 and 1
- Relationship shape: S-shaped (logistic curve)
- Coefficients: Expressed as log-odds
- Estimation method: Maximum Likelihood Estimation
- Use when: Dependent variable is binary (Yes/No)

Interaction Effects

An interaction effect occurs when the effect of one variable on the outcome depends on the value of another variable.

The diagram shows the equation $Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \epsilon$ with labels above each term: "Intercept" for β_0 , "Main Effects" for $\beta_1 x_1 + \beta_2 x_2$, "Interaction" for $\beta_3 x_1 x_2$, and "Error Term" for ϵ . Brackets group these terms. Below the equation, two lists of terms are shown with lines pointing to the coefficients in the equation. The first list, "Slope coefficients", "Effect sizes", and "Slope estimates", has lines pointing to β_1 , β_2 , and β_3 . The second list, "Covariates" and "Predictor variables", has lines pointing to x_1 , x_2 , and $x_1 x_2$.

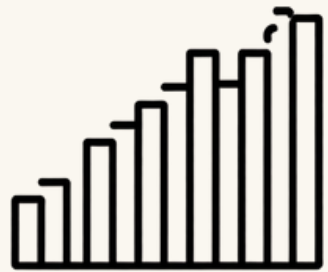
$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \epsilon,$$

- Slope coefficients
- Effect sizes
- Slope estimates
- Covariates
- Predictor variables

$$Y (\text{Wages}) = \beta_0 + \beta_1 \text{PD} + \beta_2 \text{Center} + \beta_3 (\text{PD} \times \text{Center}) + \varepsilon$$

- Without interaction: the effect of professional development (PD) on wages is assumed constant across all provider types.
- With interaction: the effect of PD on wages is allowed to vary by provider type.
- The interaction term (β_3) tests whether the effect differs by group.
- Example: Does professional development increase wages more for center-based providers than for home-based providers? The interaction term $\text{PD} \times \text{Center}$ directly tests this question, a significant β_3 indicates the effect of PD on wages is not the same for both provider types.

Summary



Model Selection

- Choose your regression model based on the dependent variable type:
 - use linear regression for continuous outcomes and logistic regression for binary outcomes.



Statistical Control

- Statistical control clarifies relationships by isolating the variable of interest, but regression alone cannot prove causality, unmeasured confounders may still bias results.



Interpretation

- Always interpret coefficients in the context of the full model and the reference group.
- Logistic regression outputs are in log-odds; convert to odds ratios or predicted probabilities for clarity.

In-Class Activity - Small Group Exercise

Instructions

Form groups of 4. Using Tables 1 and 2 from Bressane et al. (2024), discuss the following questions with your group for 10 minutes. Be prepared to share your key takeaways with the class.

- Which correlation in Table 1 surprised you the most? Why did it stand out? What does it suggest about the relationships between the variables in the study?
- Why did the authors choose to include interaction terms in their linear regression model? What theoretical or empirical reasoning supports this decision?
- How might the interpretation of the results change if interaction terms had not been included? What risks or limitations would arise from omitting them?

Next Class

- Data Visualization and Rmarkdown
- Short quiz #2

Questions?

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15-minute break